Sourav Kumar

+91-9234395033 | kumarsourav2401@gmail.com | Gaya, Bihar, India

linkedin.com/in/sourav2401 | github.com/sourav349 | souravk-portfolio.vercel.app | codolio.com/profile/sourav2401

EDUCATION

Jaypee University of Engineering and Technology **Aug 2023 – May 2027**
Bachelor of Technology in Computer Science and Engineering ***[CGPA: 7.3]*** *Guna, Madhya Pradesh*

TECHNICAL SKILLS

- **Languages:** C++, Python, Java, Golang, SQL
- **Data Analytics:** Excel, Power BI, Data Cleaning, Data Validation, EDA, Data Visualization
- **Libraries & Tools:** React.js, FastAPI, RestAPI, Streamlit, NumPy, Pandas, Scikit-learn, Docker, AWS, CI/CD
- **AI / ML:** CNNs, Image Processing, NLP, LLMs, RAG, Fine-tuning, PyTorch
- **Coursework:** DSA, OOPs, DBMS, Machine Learning, Operating System, System Design, SDLC
  Solved **950+** Data Structures and Algorithms problems on **LeetCode**, **GFG**, **HackerRank**, and **CodeChef**.

INTERNSHIP EXPERIENCE

Instruments Research & Development Establishment (IRDE–DRDO) **May 2025 – July 2025**
Machine Learning / Image Processing Intern *Dehradun, India*

- Implemented **Non-Uniformity Correction (NUC)** methods and **LSC-CNN** to enhance **infrared image quality** for **thermal imaging applications**.
- Developed and trained **CNN** and **Deep Filter Neural Networks** for **infrared image denoising and correction**.
- Designed **deep learning workflows** in **PyTorch** for **defense-focused imaging** and **restoration tasks**.

Collegedunia.com **April 2026 – June 2026**
Data Researcher Intern *Remote*

- Conducted **data research**, collection, and validation to ensure accuracy and consistency across educational datasets.
- Utilized **Python** and **Microsoft Excel** for data cleaning, analysis, and report generation.
- Supported data-driven decision-making by preparing structured reports and maintaining high-quality data records.

PROJECTS

E-commerce Conversion Prediction & Personalized Campaign Optimization | GitHub Sep 2026 – Present

- Built a **conversion propensity model** to estimate purchase likelihood from customer behavioral signals, including product views, clicks, cart activity, device type, and user history.
- Performed **EDA and funnel analysis** to identify purchase patterns, customer behavior, and key conversion drivers.
- Used **SQL and Python** for data extraction, cleaning, aggregation, and session-level analysis.
- Compared **Logistic Regression, Random Forest, and XGBoost** using ROC-AUC, PR-AUC, precision, and recall.
- Performed **feature engineering and customer segmentation** into low, medium, and high purchase-intent groups.
- Designed a **campaign strategy** to prioritize customers based on purchase propensity.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, XGBoost, SQL, Matplotlib, SHAP, Streamlit

AI-Powered Smart Product Pricing & Visual Attribute Extraction | GitHub Feb 2026 – Present

- Built a **multi-modal AI system** for product price prediction using textual, visual, and structured product attributes.
- Developed an **OCR-based entity extraction pipeline** to extract like weight, volume and dimensions from product images.
- Implemented **LightGBM-based pricing models** with feature fusion from NLP, image features, and extracted entities.
- **Tech Stack:** Python, LightGBM, OCR, OpenCV, PyTorch, Transformers, FastAPI, React.js, Docker

CERTIFICATIONS

Problem Solving (Basic) — HackerRank Issued Jul 2024

- Skills: **Problem Solving**, Data Structures, Front-End Development

ACHIEVEMENTS & LEADERSHIP

- Qualified to **3rd-phase as solo participant in RideHack'24 at JIIT** (Innovation-Startup track).
- **Qualified for Smart India Hackathon (SIH) Internal 2025** at JUET.
- **Coordinator** at **Rospinot and Mozilla clubs**; led initiatives and promoted tech community collaboration.
- **Hacktoberfest 2025 Supercontributor** with **6+ accepted open-source PRs/MRs**.